

AI-Driven Health Support: Development of a Cancer Risk Assessment and General Health Chatbot Using NLP and Machine Learning

by

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This thesis has been submitted in partial fulfillment for the degree of Bachelor of Science in Software Development

in the

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Declaration of Authorship

This report, AI-Driven Health Support: Development of a Cancer Risk Assessment and General Health Chatbot Using NLP and Machine Learning, is submitted in partial fulfillment of the requirements of Bachelor of Science in Software Development at

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- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this project report is entirely my own work.
- I have acknowledged all main sources of help.
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Signed: Betul Kantaroglu

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Date: 20/12/2024`

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MUNSTER TECHNOLOGICAL UNIVERSITY CORK

Abstract

Faculty of Engineering and Science
Department of Computer Science

Bachelor of Science

by Betul Kantaroglu

Abstract

Advancements in medical technology have steered healthcare in an innovative direction. It has revolutionized patient care like never before.

This thesis focuses on the development of a personalized webpage where patients are able to login/sign up using secure methods, access their dashboard, chat with any of the two chatbots that are available. One of these chatbots is a cancer risk assessment chatbot that uses machine learning models trained on symptom-based datasets. It lets patients choose the type of cancer they would like to pick from the following cancers:

- Breast cancer
- Lung cancer
- Prostate cancer
- Skin cancer
- Cervical cancer

The bot then prompts the users with questions and returns a risk score of low, medium or high based on the symptoms inputted by users. Each risk score will be recorded on to the users dashboard, where they will be able to access them to compare.

Meanwhile, the second chatbot answers general health-related questions or concerns. For instance, if a user mentions 'I have a headache" the model will follow up with more questions to grasp the problem better and will provide non-diagnostic tips. The user can additionally ask any questions they may have on health or lifestyle, even if they are not experiencing any symptoms. This model uses NLP to enhance readability and user engagement. The aim is to educate users on any issues or questions they may have in real-time.

There were a few significant challenges in this project that had to be addressed, for example, the lack of symptom-based datasets available ensuring compatibility across multiple technologies, including NLP frameworks, web applications, and databases. These challenges were addressed through techniques such as synthetic data generation and iterative testing. The functional requirements of the system include real-time symptom evaluation, personalized health advice generation, and an interface that is both accessible

(with features like high contrast themes and font size adjustment) and easy to use for everyone regardless of their technical skills.

The project is aligned with the ongoing trends in artificial intelligence. Upon completion, the platform will serve as an accessible and free platform to increase patient awareness on cancer related symptoms and general health.

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Abbreviations

LAH **L**ist **A**bbreviations **H**ere

For/Dedicated to/To my...

Chapter 1

Introduction

It is no doubt that advancements in medical technology have saved and improved the lives of many. Artificial intelligence is a hot topic in technology at the moment and has opened many doors in medicine by offering innovative solutions to longstanding challenges. This project uses AI to address two critical areas:

- Early detection of cancer educating patients on symptoms
- Providing general health advice

By integrating web development, database management, and AI-driven analysis, the project aims to create a dual-chatbot platform that allows patients to gain insight on general health related topics and assess their cancer risk based on the symptoms they provide to the chatbot.

1.1 Motivation

The motivation behind creating a cancer risk assessing project comes from the urgent need to improve early detection rate for cancer. Cancer has been the leading cause of death for over 75 years. In 2022 alone, cancer was responsible for over 9.7 million deaths.

Despite advancements in technology, many patients are not diagnosed early enough, significantly reducing their chance of survival.

I have had firsthand experience with the negative impact delayed cancer diagnosis can have on individuals and their loved ones. I have always wondered how many lives could have been saved if patients were diagnosed and treated much earlier, especially in regions or communities where access to private medical care may be limited.

With the rapid advancement of AI, there is a valuable opportunity to leverage this technology to assist in cancer detection based on patient's individual symptoms. I believe that using AI can assist in early diagnosis which can help to save lives. This project will allow users to asses their risk on the following types of cancer:

- Breast cancer
- Lung cancer
- Cervical cancer
- Prostate cancer
- Skin cancer

The motivation behind creating a second chatbot which deals with general health is to provide individuals with a tool that will answer any questions they may have regarding health. This tool will be useful for everyone but especially to those who may have questions they need answers or advice in real time. I feel that this tool will be useful to those who may have limited resources or need to wait a long period of time to be able to speak to a doctor.

The chatbot will be able to analyse the symptoms the patient is experiencing, ask follow up questions depending on their situation and provide personalised non-diagnostic advice regarding their symptoms.

1.2 Contribution

The main contributions of project focused on using the computer science skills and methodologies I have acquired during my studies.

Web Development and UI/UX Design: This project will use Flask to provide a user-friendly interface ,it will guide patients to login/sign up using secure methods, once logged in they can chat with the general health chabot or use the AI symptom checker to analyze their symptoms and get a cancer detection analysis on the type of cancer they wish. These scores will be recorded on their dashboard. They will be able to revisit the website and view their previous results, this will allow them to track symptom changes over time.

Back-End Development and Authentication Systems: The back-end for this project will focus on secure user authentication and data management. This authentication 3

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system will ensure that every patient will have a personalized dashboard where they can view their center risk analysis and retest if they wish. This will be created using Flask.

Database management and Data Logging: My database skills were acquired through modules such as COMP6041- Introduction To Databases. SOFT7022- Database Design and COMP7037 NoSQL Data Architectures. I also had hands-on experience through my 3rd year internship with McKesson, where I worked heavily with databases.

A database system will be established to store and manage user data. This data will include things such as login credentials, symptom entries and cancer risk scores. Storing each users information in a database allows them to revisit their past results and track changes in their results in real-time.

Machine Learning for Symptom Analysis: The AI cancer detector will be implemented using machine learning principles and techniques learned from my COMP8043 Machine Learning module from 4th year and SOFT8032 - Programming for Data Analytics from 3rd year. Data preprocessing, model deployment and integration will be applied into a web application. Using pre-trained machine learning models, Python libraries like scikit-learn or TensorFlow will be implemented within Flask to ensure smooth model execution, allowing real-time predictions on symptom data. year.

General Health Chatbot Development: The general health chatbot will employ state-of-the-art NLP techniques, including pre-trained models like BERT or GPT, to understand user inputs and provide personalized responses. This chatbot will enhance accessibility to non-diagnostic health advice, offering real-time symptom assessments and actionable recommendations based on user inputs. These skills will be acquired through official documentations, YouTube and courses on educational platforms like Udemy.

These modules have made it possible to develop this project.

1.3 Structure of This Document

This document is structured as follows: - **Chapter 1** Provides an introduction to the project, detailing the envisioned final state this project aims to achieve. - **Chapter 2** Chapter : Provides a thorough rundown of the thematic area surrounding this project, including health informatics, data analytics, artificial intelligence, and machine learning. It examines the current state of the art in AI-driven cancer risk assessment and general health chatbots, highlighting key trends, methodologies, and gaps that currently exist. Companies, conferences and platforms that correlate to this project are also discussed. Additionally, The positioning of this project within the broader field of computer science is established in this chapter, discussing its relationship to topics like software development, machine learning, cybersecurity, database management, and human-computer interaction. The strategy for designing and implementing this project is shaped by the insights gained from this chapter.

- **Chapter 3** Discusses the problem this project aims to address:

  - Lack of accessible tools for cancer risk detection.
  - A reliable tool that provides health advice in real time.

It discusses why early cancer detection is so important and the struggle people may face when trying to get health advice instantaneously during difficult times. This chapter also explores the problems associated with cancer diagnosis and general health advice such as accurate symptom-based analysis and natural language understanding. It delves deeper into the objectives of the project, including the development of AI-driven chatbots, a user-friendly interface and secure data management. Finally, the chapter discusses the functional and non-functional requirements required for this project.

-**Chapter 4** : This chapter discusses the planned methodology for developing the platform. It explains the technologies, architecture, and methodologies that will be used.

-**Chapter **??****: The key findings, challenges, solutions and progress made during this phase are summarised here along with future work.

Chapter 2

Background

This chapter includes an overview of the thematic area in which this project resides and a review of the current work research that has been done on this topic. The focus is on identifying existing work and research in relation to AI-driven cancer detection chatbots and general health chatbots, understanding the computational methods employed, and contextualizing this project within the broader field of Computer Science.

1. The core topic of this project is the development of a web-based platform for AI driven cancer risk detection and an AI-driven general medical advice chatbots. This project aims to provide patients a web interface to log in/sign up, respond to symptom-based questions provided by a cancer risk assessing chatbot and also track changes in their symptoms over time, which will then produce an updated risk assessment. If they do not wish to assess their risks on various different types of cancer, they are able to chat with a general health chatbot, which will provide patients with general tips and advice on general health queries.
2. The core areas this project falls under are:
    - Health Informatics: Analysing and managing health-related data for the risk of cancer and general health queries through computational techniques. This includes leveraging health-related knowledge bases and integrating insights for accurate chatbot responses.
    - Data Analytics: Applying techniques for data preprocessing, visualization and examining trends to get an understanding of symptom progressions and risk levels.
    - Artificial Intelligence And Machine Learning: Utilizing AI-driven models to analyse the symptoms of each user and computing their cancer risk and implementing NLP techniques such as BERT and GPT for the general health chatbot, these techniques allow queries to be processed and responded to despite grammar errors, varying slang and spelling errors.

- Web-Based Application: Developing a user-friendly web interface for patients to log in, view their personal dashboard- which consists of their previous risk assessments, and interact with either of the chatbots to compute a new risk assessment based on their current symptoms or to inquire about general health related questions.
- Service Providers: Providing a digital service that aims to improve patient awareness for cancer symptoms and additionally provide general insight and advice on everyday health questions.

3. The main areas of Computer Science this project falls under are:

   - Software Development: This project requires the development of a user-friendly web interface. This is crucial as it ensures every patient is able to use the platform regardless of their technical background. This interface will serve as the main point of interaction for users, ensuring they can:
     - Login/ Sign up securely and manage their profiles.
     - Obtain a cancer risk score by selecting the specific type of cancer they wish to assess (e.g., lung cancer, breast cancer, cervical cancer, prostate cancer, skin cancer) and answering relevant questions through interaction with the **Cancer Risk Chatbot**
     - View their past cancer risk scores on their dashboard, allowing them to track changes in scores over time.
     - Ask **general Health Chatbot**. any questions they may have about general health or receive non-diagnostic health advice.

       The software development process involves integrating pre-trained AI models to analyse patients' inputted symptoms in real time and generate personalized cancer risk assessment scores through the Cancer Risk Chatbot. This feature requires a database to store user information such as login information and previous risk scores. This allows users to view their previous risk scores and track their health changes over time. Similarly, the General Health Chatbot uses natural language processing models to answer general health-related queries and provide non-diagnostic tips to users.

       Additionally, the development process focuses on integrating the AI models with the database seamlessly and ensuring that both chatbots operate efficiently and accurately.

- Cybersecurity: Cybersecurity is crucial as the system deals with sensitive user information. To protect user data against unauthorized access, secure handling and processing of user data must be ensured.

  In order to achieve this, the platform will implement several key cybersecurity measures. These measures include the use of end-to-end encryption for data in transit and at rest. The use of end-to-end encryption guarantees that all user data remains confidential during communication with the chatbot, storage in the database, and retrieval processes. Secure Authentication Mechanisms like multi-factor authentication (MFA) and password hashing will be adopted in order to verify the identity of users and prevent unauthorized access.

  Security will be further enhanced by adopting role-based access control (RBAC). This will ensure that sensitive information is only accessed by authorized users and systems. By adopting this feature, users will be categorized into roles such as:

  * Patients: These users can only access their risk scores.
  * Administrators: These users will have the necessary permissions to manage the system, monitor usage etc..

  The platform will follow secure coding practices in order to lessen the risk of common vulnerabilities like, SQL injection and cross-site scripting (XSS). Input validation will be implemented to guarantee that all inputs by users are safe to process.

  Lastly, basic logging and monitoring will be used to track any unusual activity that may occur and track login attempts. In order to reduce the possibilities of any data loss, regular database backups will be scheduled. The implementation of these cybersecurity measures will ensure the safe handling of data while simultaneously exhibiting a powerful understanding of security principles.

- Machine Learning: Machine learning will be the main component of this project. This is another area of Computer Science this project falls under. The aim is to develop two AI-driven chatbots. One of these chatbots will handle general health queries. This chatbot will provide general tips and advice. The other chatbot will assist patients in assessing their level of risk for different types of cancer. The aim is to develop a chatbot that prompts patients to select a specific type of cancer they would like to assess their risk on. Based on their selection, they will be asked symptom based

questions about the cancer they have chosen. The chatbot will then analyse the responses inputted by the patient and determine a risk score.

1. Cancer Risk Assessment Chatbot: Key Features of the Chatbot: 1. Cancer Types that will be assessed:
   * Lung Cancer
   * Breast Cancer
   * Prostate Cancer
   * Skin Cancer
   * Cervical Cancer
   * patients will be prompted one of the cancers listed above.
2. Questioning: Once the user has selected the cancer type they would like to assess their risk on, the chatbot will ask a series of questions that will be specific to the cancer selected.
   * For example, for skin cancer, users may be prompted to answers symptom based questions such as:
     · ”Is there a change in shape, colour or size to an existing spot?”
     · ”Do you have a sore that doesn’t heal?”
     · ”Do you have a new lump that may be red, slow growing or shiny?”
   * For lung cancer, questions might focus on coughing up blood, chest pain, smoking history and exposure to pollutants.
3. AI-driven Analysis:
   * The responses provided by the user will be analysed by the chatbot using pre-trained machine learning models specific to the selected cancer type. These models will be trained on datasets that contains historical symptom data, patient demographics, and known risk factors
   * Once analysed, it will return a risk assessment score such as:
   * Low Risk
   * Moderate Risk
   * High Risk
4. User-Friendly Design:

– The chatbot will simulate a natural conversation. This is done to make the experience more interactive and engaging, which will help patients feel

they are discussing their symptoms with a virtual assistant rather than just filling out a questionnaire.

- Recommendations and Next Steps: The chatbot will recommend the next steps the patient should take based on the risk assessment. These recommendations may include consulting a healthcare provider or undergoing specific diagnosis tests. In order to avoid replacing professional medical advice, the advice will remain general and non-diagnostic.

  2. General Medical Advice Chatbot:

- Patients will be able to ask for tips and advice for health issues unrelated to cancer.

  Example interaction:

  * User: ”Hello, I have a headache, what should I do?’
  * Chatbot: ”Hey there! Try staying hydrated, try to avoid screen time and make sure you get plenty of rest in a quiet dark room. If this issue persists please consider consulting a medical professional.”
  * User: ”Hello, do you have any advice for a sore throat?”
  * Chatbot: ”Ensure you get plenty of fluids, gargle with warm salt water, consider lozenges, if this issue persists please consider consulting a medical professional.

- This functionality will utilize pre-defined responses based on common health scenarios. This will be augmented by natural language processing (NLP) models. This will ensure that the responses will be in a conversational tone.

  Providing users with two chatbots increases engagement, as patients are likely to have health related questions that may not be just cancer related.

- Database Management: Database management is another crucial area of computer science area this project addresses. This aspect of the project is crucial in order to ensure efficient storage, retrieval and secure handling of patient data and chatbot interactions.

- Role of the database in this project:

  * 1. User Authentication and Profile Management:
    - The database will safely store user login information.
    - it will keep track of patient profiles, these profiles will include necessary information required for accurate risk assessments.
  * 2.Symptom and interaction Logging:

- The system will log the inputs provided by the patients for both cancer risk assessments and for general medical advice queries.
- These recorded logs will enable patients to track changes in their symptoms over time and revisit their previous interactions.

* Previous Risk Assessments:
  - Each patients relevant cancer risk assessment results will be stored, allowing patients to view their previous assessments, this will enable them to compare results over multiple assessments.

* Managing questions and responses:
  - A repository of questions for the chatbot, categorized by cancer type or general health topics, will be stored in the database.
  - The database system will contain the pre-defined responses or suggestions for the general medical queries. This can be updated dynamically as needed.
  - This setup will promote flexibility in the chatbot, allowing for updates without significant disruptions.

* Real-Time Data Integration:
  - The chatbots will continuously be in communication with the database in real-time, ensuring that each patient input is logged instantly without any delay and therefore respond with instant cancer risk assessments or a general medical advice.
  - This features promotes a seamless user experience as it will improve the responsiveness of the chatbot.

* Security and Privacy
  - Security will be further enhanced by adopting role-based access control (RBAC). This will ensure that sensitive information is only accessed by authorized users and systems. By adopting this feature, users will be categorized into roles such
  - Patients: These users can only access their own risk-assessments.
  - Administrators: These users will have the necessary permissions to manage the system, monitor usage etc..

* Human-Computer Interaction (HCI) and UI/UX Design
  - Human-Computer Interaction (HCI) and User Interface/User Experience (UI/UX) are crucial to Computer Science. These

disciplines focus on making the systems easy to use and interactive to users regardless of a users technical background.

· User-centred Design:

· The main goal of the UI/UX in this project is to maintain a high quality and professional UI while simultaneously making it efficient to use.

· Colour schemes, typography and layout will be chosen carefully in order to establish a professional and calming interface.

· Clear Navigation:

· As mentioned, the system aims to be straightforward. The functionalities of the web page will be clearly labelled.

· Patients will be able to:

· Log in/Sign up

· Choose a chatbot, i.e. the general medical advice chatbot or cancer risk assessment chatbot.

· Interact with the chatbot · View

their past risk assessments.

· Accessibility Features:

· In order to ensure that patients with disabilities or impairments are also able to use the platform with ease, features such as adjustable font sizes and high-contrast themes will also be implemented.

· Feedback:

· Immediate feedback will be provided to patients. If there are invalid input formats, users will be prompted to try again.

2.1 A Review of AI-Driven Cancer Risk Assessment and Chatbots

- **The top 5 International Conferences:**

  – 1. International Conferences:

1. **Conference on Neural Information Processing Systems (NeurIPS):** This conference focuses on the advancements in AI and machine learning technology, which are significant in the development of AI models for providing both cancer risk assessment and general health advice.

2. **International Conference on Machine Learning (ICML):** ICML is recognised globally as for publishing breakthroughs in machine learning in areas such as artificial intelligence, statistics and data science. Other areas of interest such as machine vision, computational biology, speech recognition, and robotics are also discussed.

3. **IEEE International Conference on Healthcare Informatics (ICHI):** ICHI focuses on utilizing computer science, data science and informatics principle along with information technology, communication science and technology to tackle problems in areas such as medicine, public health, healthcare and general health.

4. **American Medical Informatics Association (AMIA) Symposium**: The AMIA Annual Symposium stands out as the event, for exploring advancements in biomedical and health informatics research and application. The conference brings together many professions that are relevant to this project:

   · Data Scientists

   · Computer Scientists

   · AI developers

   · Physicians

   · Nurses

5. **Artificial Intelligence in Medicine (AIME) Conference**: This conference focuses on practical applications of AI advancements and applications in medicine, latest trends and relevant research.

   Areas discussed in the conference include but are not limited to:

   · Machine learning and big data analytics

· Natural language processing

· AI for biomedical imaging

The top 5 Journals:

1. **Artificial Intelligence in Medicine (Elsevier): The journal, Artificial Intelligence in Medicine delves into using AI methods to tackle issues in the field. [1].**

   Example articles that are relevant to this project:

   · Breast cancer detection using artificial intelligence techniques: A systematic literature review [2]

   · AI in medical diagnosis: AI prediction human judgment [3]

2. **The Lancet Digital Health: The Lancet Digital Health publishes transformational and important research on the use of digital technology in clinical medicine, public health and global health. [4] ∗ Example articles that are relevant to this project:**

   · Generative artificial intelligence and ethical considerations in health care: a scoping review and ethics checklist [5]

   · A future role for health applications of large language models depends on regulators enforcing safety standards [6]

3. **Nature Medicine: Nature Medicine publish studies on science and technology. [7]**

   Example articles that are relevant to this project:

   · AI in health and medicine [8]

   · Large language models in medicine [9]

4. **Journal of Biomedical Informatics: Journal of Biomedical Informatics publishes articles and research conducted on AI applications in healthcare.**

· Example articles that are relevant to this project:

· Machine Learning based Intelligent System for Breast Cancer Prediction [10]

· Prostate cancer prognosis using machine learning: A critical review of survival analysis methods[11]

5. **IEEE Transactions on Biomedical Engineering: Discusses Artificial intelligence and data analysis in the field of biomedicine.**

   Example articles that are relevant to this project:

   · Cancer Prediction using Machine Learning Prediction [12]

   · Cancer Detection and Analysis Using Machine Learning methods[13]

The top 5 Books:

1. **"Deep Learning for Medical Image Analysis" by Zhou et al. (2020) Explanation:**

   – Although this books main focus is on medical image analysis, it is an excellent source for grasping the concepts of deep learning and machine learning and how they are applied in healthcare.

   – Concepts such as:

   (a) Preprocessing Data

   (b) Training model

   (c) Evaluating performance

   are highly relevant for the design and implementation of the AI models that will be required for this project.

2. **"Python Machine Learning" by Sebastian Raschka and Vahid Mirjalili (4th Edition, 2022) Explanation:**

- This book provides an extensive exploration of machine learning techniques using python. It provides insight on libraries like Scikit-learn, TensorFlow, and PyTorch. Additionally, it provides advice on:

  (a) Machine learning algorithms

  (b) Model selection

  (c) Hyperparameter tuning

  (d) Evaluation

  (e) Deploying trained models into web based systems

- This text will play a crucial role during the development phase of the cancer risk assessment chatbot as well as in creating the general health chatbot.

3. **"Artificial Intelligence and Machine Learning in Healthcare" by Ankur Saxena (2021) Explanation:**

   - This book explores uses of AI and machine learning in the healthcare industry, covering areas like patient data analysis, predictive models, and ethical considerations. It offers advice and knowledge on areas such as:

     (a) Challenges (e.g., data privacy and algorithm bias)

     (b) Developing AI tools that preserve accuracy and reliability

     (c) AI implementation in healthcare

   - This book offers valuable knowledge into developing technical precision while also maintaining a user-friendly design. This enables optimum chatbot performance and usability.

Top 5 Companies/Organizations Potentially Interested in the Product:

1. **Hospitals and Healthcare Providers:**

- **Why they could be interested:** Healthcare facilities and hospitals are constantly looking for innovative tools to enhance patient outcomes and early diagnosis. Their services would benefit greatly from the addition of a chatbot for cancer risk assessment that offers rapid, symptom-based analysis, assisting in early detection and reducing the strain on diagnostic teams. A chatbot that answers questions on general health queries would also be beneficial for patients to get tips and advice efficiently and in real-time. This could help reduce the workload of professionals.
- **Potential Use:** Healthcare facilities and companies could incorporate the chatbots into their portals or mobile applications to offer patients a screening process before they seek medical advice from a healthcare provider.

2. **Health Insurance Companies:**

- **Why they could be interested:** Health insurance companies encourage preventative care through tools like a cancer risk assessment chatbot to detect cancer early in an effort to promote lower overall expenses. Additionally, a general health chatbot can offer guidance on handling health concerns without needing to see a doctor in person, thereby reducing insurance claims.
- **Market use case:** In order to encourage users to engage in preventative health management, the chatbots could be added into insurance apps as a value-added service.

3. **Telemedicine Platforms (e.g., Teladoc Health, Amwell):**

- **Why they could be interested:** Telemedicine services are searching for ways to enhance virtual care experiences. Before patients connect with doctors, the chatbot can provide symptom evaluation. This would improve the effectiveness of online consultations. For less urgent problems, the general health chatbot could offer patients guidance. This, in return, may assist in lowering the number of needless consultations.
- **Potential Use:** Chatbots have the potential to be incorporated into telemedicine systems in order to simplify communication between patients and doctors, functioning as a support for assessing symptoms.

4. **AI and Health Tech Companies (e.g., IBM Watson Health, Google**

Health):

- **Why they could be interested:** In the realm of healthcare innovation are organizations such as IBM Watson Health and Google Health. These companies are focused on addressing healthcare challenges effectively. This project's objective aligns with their mission of developing AI-driven tools for better health results.
- **Potential Use:** These companies may either take on the project for their research and development efforts or could collaborate to improve and scale it for widespread use.

5. **Public Health Organizations (e.g., World Health Organization, Centers for Disease Control and Prevention):**

- **Why they could be interested:** Public Health Organizations' main goal is to prevent global diseases and to improve health results. The chatbot developed for cancer risk assessment may assist in promoting detection efforts. The general health queries could increase overall awareness. These chatbots adopted globally could benefit communities and households where healthcare resources may be restricted.
- **Potential Use:** These companies could use these chatbots in their outreach efforts to inform the community and offer health evaluations in isolated regions.

Top 5 Wiki/Forums/Blogs/Youtube Channels Related to the Project:

1. **Stack Overflow (Forum):** Stack Overflow is a valuable forum for this project. It is a platform that is used globally by developers and computer scientists to discuss problems they may encounter and possible solutions for them. Since developers provide practical advice on all areas of software development, it will be beneficial if a problem is encountered during the development phase of this project.
2. **Towards Data Science (Blog):** Towards Data Science is a blog that focuses heavily on topics such as machine learning, AI, and data science. It offers users tutorials, case studies, and provides real-world applications of AI technologies in various fields, including healthcare. These articles will provide meaningful insight as they relate directly to the topic this project will be based on.

3. **Kaggle Forums and Competitions (Forum):** Both data and computer scientists converse and share practical tips and advice on AI techniques, share datasets, and compete in challenges. Kaggle serves as an excellent resource for learning about machine learning and for obtaining required datasets for the project.

4. **YouTube Channels - ”Python Engineer” and ”StatQuest with Josh Starmer”:** YouTube channels like ”Python Engineer” and ”StatQuest with Josh Starmer” upload clear and detailed tutorials on machine learning, AI, and Python for all levels. They offer step-by-step tutorials on Flask, TensorFlow, and chatbot development. StatQuest simplifies complex machine learning concepts, which will aid in both implementing and the theoretical aspects of chatbots required for this project.

5. **Reddit Communities – r/Python, r/MachineLearning and r/HealthcareAI:** Reddit communities are a great way of conversing with other developers. Subreddits such as r/Python, r/MachineLearning, and r/HealthcareAI consist of computer scientists and enthusiasts where problems, solutions, recent advancements, articles, and projects are shared and discussed. Subreddits are interactive and allow for feedback and suggestions by other users.

2.2 Related Work

The use of chatbots and artificial intelligence (AI) has led to significant advancements in medicine and health. This section reviews related works to position the current project within the broader context of existing research. Key trends, challenges, and gaps are identified to demonstrate how this project contributes to the field.

2.2.1 Applications of Chatbots in Oncology

Study Objective: JAMA Oncology [14] analyzed the effectiveness of chatbots in addressing cancer-related queries. Four chatbots (ChatGPT, Perplexity, ChatSonic, and Bing AI) were tested with five frequently asked questions across different cancer types (skin, colon, prostate, lung, breast).

Key Findings:

- The chatbots averaged an accuracy score between 4 and 5 (on a scale of 1 to 5), indicating high-quality information.

- The understandability score was between 50% to 72.7%, which indicated moderate readability.
- Actionability scores performed poorly and stored between 0% and 40%, signalling that the chatbots were not able to offer personalised advice properly.
- The length of the responses were between 91 and 146 which could be a sign that the they lack depth.

Limitations: Responses consisted of medical terminology and were not easily actionable. possibly affecting user satisfaction.

Relevance to This Project: The findings of this review highlight the importance of a chatbot that offers readability, personalisation while still maintaining accuracy. This project aims to address work through these limitations by using symptom-based datasets and tailoring responses through NLP customization.

2.2.2 BERT-Based Medical Chatbots

Study Objective: Babu et al. [15] aims to overcome limitations in conventional machine learning-based chatbots by introducing a BERT-based chatbot, which aims to understand more complex medical queries.

Key Findings:

- Achieved an F1 score of 98% for its ability to understand and respond to complex queries.
- Did not have an issue processing inputs that may have grammar errors, slang, and misspellings.
- Supported coherent multi-turn conversations, improving user satisfaction and trust.

Limitations: High computational requirements for fine-tuning and deployment.

Relevance to This Project: This study guides the development of the general health chatbot in this project by displaying the importance of fine-tuning pre-trained models like BERT for healthcare-specific tasks.

The chatbots performance will be evaluated using metrics such as F1 score and precision.

2.2.3 AI for Cancer Education and Awareness

Study Objective: Didier [16] reviewed the role of AI in educating cancer patients.

Key Findings:

- 89% of patients reported gaining significant knowledge from chatbots.
- 78% of users (average age 68) found chatbots easy to use without assistance.
- Highlighted concerns about dataset biases, especially the exclusion of minority populations.

Limitations: Potential for bias in training datasets, requiring regular updates and feedback loops.

Relevance to This Project: This review showcases the importance of dataset diversity and user feedback loops.

This project will focus on data diversity and use synthetic data generation techniques to avoid data bias. The system will developed in increments to refine and adjust as needed.

2.2.4 General Trends in Healthcare Chatbots

Study Objective: Infobip [17] provided an industry overview of chatbot usage in healthcare.

Key Statistics:

- 74% of users preferred chatbots for simple queries.
- 52% of patients in the U.S. use chatbots for healthcare information.
- There are chatbots currently available that offer features such as medication reminders, chronic disease management, and emergency assistance.

Limitations: Limited focus on symptom assessment or diagnostic accuracy.

Relevance to This Project: This article reports on the growing acceptance of chatbots in healthcare. This project aims to expand on these functionalities by integrating symptom assessment and personalized health tips.

2.2.5 Empathy and Accuracy in AI Chatbots

Study Objective: A study by the National Cancer Institute [18] highlighted the need for regular updates in cancer-related AI models.

Key Findings:

- Chatbots responded in an empathetic manner however, responses often lacked accuracy due to outdated datasets.
- Regular updates to AI models were suggested to ensure that the models are upto-date with the latest health data.

Relevance to This Project: This study underlines the importance of working with up-to-date datasets. This project will aim to incorporate feedback and will use up-todate datasets.

The reviews and articles discussed above highlights the important challenges, statistics and benefits regarding the matters of this project. It offers insightful knowledge on what is currently available and how it can be improved. These key findings will help shape the project in the developmental phase.

Chapter 3

Problem-AI-Driven Health and Cancer Risk Assessment Chatbot

3.1 Introduction

The merging of artificial intelligence with healthcare has created vast possibilities to tackle persistent challenges in patient care and diagnosis. Among the most pressing is the delay in cancer detection and treatment. Moreover, there is also a need for quick and reliable tools that will provide patients with general guidance on basic health queries, especially in times of need when professional help may not be available immediately.

This project aims to address this challenge by developing two AI-powered solutions: a Cancer Risk Assessment Chatbot and a General Health Query Chatbot. By combining natural language processing (NLP) with AI, the goal is to provide users with a simple yet effective tool to help manage and query about the current symptoms they may be dealing with.

3.2 Problem Statement

The primary problem this project seeks to address is the lack of accessible, personalized, and symptom-based tools for general health advice and early cancer risk detection. By addressing this problem, patients will be given a platform where they can gain insight into their health in times of need.

3.2.1 Cancer Diagnosis

Early diagnosis can save lives; however, an early diagnosis is not always possible. Late cancer diagnosis could be due to several factors such as:

- Misdiagnosis
- Lack of financial resources
- Long waiting times for consultation
- Patients may ignore their symptoms and not see a professional early enough
- Patients may be uneducated about the symptoms they are experiencing.

According to a recent article published by City of Hope [19], it is reported by the organization that even a one-month delay in diagnosis could increase the risk of fatality for cancer patients.

The more time cancer patients are left without treatment, the more likely the cancer cells are to grow and spread to other parts of the body. This is called metastasis, and it makes the cancer more difficult to treat effectively. The proof of this is evident in the survival rates of cancers, with cancer metastasis being the main reason behind cancer-related deaths.

The chart below demonstrates the drastic difference in the five-year survival rate for patients diagnosed with different types of cancer, depending on when they were diagnosed:

- **Localized:** Cancer has not spread to lymph nodes or nearby tissues.
- **Regional:** Cancer has spread to nearby lymph nodes or tissues.
- **Distant:** Cancer has metastasized to other organs or tissues.

Type	Localized	Regional	Distant
Breast cancer (female)	99.6%	86.7%	31.9%
Prostate cancer	100%	100%	36.6%
Lung cancer	63.7%	35.9%	8.9%
Colorectal cancer	91.1%	73.7%	15.7%
Melanoma	100%	74.8%	35%

Table 3.1: Five-Year Survival Rates by Diagnosis Stage

The chart above highlights the importance of early detection in order to prevent/reduce the risk of metastasis, thereby increasing the chances of survival.

3.2.2 General Health

Acquiring information about general health is also very important. According to an article released by Eurostat [20] in 2022, 55% of EU citizens aged between 16 and 74 have reported seeking medical and nutrition information online. This can be due to several factors:

- **Convenience:** Patients may find it easier to turn to the internet for quick answers.
- **Lack of financial resources:** Not everyone may have the resources to pay the high costs of consultations.
- **Unavailable immediate help:** Immediate access to a medical professional may not be available, and patients may face long waiting times.

Although individuals search for answers on the internet to better understand the problems they are experiencing and to gain insight into their conditions, this usually has the potential to backfire, becoming misleading and damaging. The internet can provide insight and advice; however, it may not always be personalized or reliable, which can be dangerous.

3.3 Key Challenges

1. **Accurate Symptom-Based Analysis:** Providing accurate assessment risk scores based purely on symptoms reported by patients may be a challenge, as there is limited availability of symptom-based datasets for different types of cancers.
2. **Natural Language Understanding:** Ensuring the general chatbot understands and processes patients' inputs, which may vary in spelling, grammar, and phrases, accurately and provides actionable responses.

3.4 Objectives

This project aims to achieve the following:

3.4.1 Chatbot Developments

- Develop a chatbot that enables users to choose the type of cancer they would like to assess, ask symptom-based questions, and provide a cancer risk score.
- Develop a general health query chatbot that offers patients non-diagnostic general health advice and answers common health-related questions, such as headaches, coughs, and dizziness.

3.4.2 User-Friendly Interface & Accessibility

- Design and develop a user-friendly interface that is efficient for every user of the platform, regardless of their technical background.
- Provide accessibility features such as contrast themes and adjustable font sizes throughout the website.

3.4.3 Secure Data Management

- Store necessary patient data such as login information and past risk scores securely in a database.
- Use measures such as encryption and secure authentication to protect sensitive user data.

3.4.4 AI Integration

- Utilize publicly available symptom-based datasets, modify or create new ones if necessary, and train and deploy AI models for cancer risk assessments.
- Incorporate natural language processing (NLP) functionalities to allow the chatbots to interact naturally with users.

3.4.5 Recommendation & Tracking

- Recommend actionable steps based on the patient's risk score and allow tracking of symptoms over time.

3.5 Functional Requirements

- **Chatbot Functionality:**
  - Cancer risk assessment chatbot that allows users to choose from cancers such as:
    - Lung cancer
    - Breast cancer
    - Skin cancer
    - Prostate cancer
    - Cervical cancer
  - General health query chatbot to provide accurate information and tips for everyday health questions.
- **User Authentication and Profile Management:**
  - Secure login and account creation.
  - Ability to view past risk assessments.
- **Symptom Logging and Analysis:**
  - Efficient storage and retrieval of user data.
- **User-Friendly Interface:**
  - Professional design with high accessibility features.

3.6 Non-Functional Requirements

- **Performance:** Ensure efficient response times and high availability.
- **Scalability:** Design for increased user capacity.
- **Security:** Use encryption and secure authentication.

- **Accessibility:** Provide accessibility features like high-contrast themes.
- **Maintainability:** Write modular, documented code.
- **Ethical and Legal Compliance:** Adhere to GDPR and HIPAA standards.
- **Usability:** Conduct usability testing to ensure user efficiency.

Chapter 4

Implementation Approach

Technologies involved

This chapter thoroughly discusses the planned approach for implementing the cancer risk assessment and general health chatbot system, more specifically the technologies, architecture, and methodologies that will be used.

4.1 Technologies Involved

1. **Programming Language** - Python: Python will be the main programming language used during this project since it has extensive libraries and frameworks for machine learning, AI, data analysis, and web development. Its versatility and the large community of developers make it a reasonable choice for development.

   Applications in the project:

   - Development of both the general health query chatbot and cancer risk assessment.
   - Integrating machine learning models.
   - Managing the backend processes.
   - Managing APIs and database interactions efficiently.
   - It also will help in reducing development time and will enhance maintainability as it has simple syntax.

2. Web Development Frameworks: Essential for developing an accessible and easy to use platform that will allow users to interact with either/both of the chatbots.
   The frameworks considered for this project:

   (a) Flask: Flask provides flexible and lightweight modular tools, which makes it suitable for choosing it as the web framework.

   Features of Flask:

- Flask offers tools to build APIs that will handle requests between the frontend and backend.
- To achieve iterative development, Flask allows for rapid prototyping.
- Enables the connection of AI models and databases through its extensions.

Applications in the project:

- Chatbot interaction: In order to send and receive data between users and the chatbot, Flask can be used to create endpoints.
- User Authentication: Flask-Login and Flask-WTF are libraries of Flask which will facilitate secure sign-up and login for users.
- Database Management: Flask's SQLAlchemy extension ensures that user data is handled efficiently by managing the connections to the database.

(b) Django: Although the goal is to use Flask throughout the project, if more robust needs arise Django may be used as an alternative.

Why Django may be used:

i. Since it is widely used for larger-scale projects, it is able to support an increase in user numbers on the platform.

ii. Its emphasis on security.

iii. Django has built-in database management features like admin panels, form validation, and ORM (Object-Relational Mapping).

3. Machine Learning and Deep Learning:

Scikit-learn: Scikit-learn is a lightweight library that consists of rich tools which will be used for building and evaluating classification models.

Applications in this project:

- Model building: Classification models such as Logistic Regression, Random Forest, and Decision Trees will be implemented to evaluate the cancer risk of users.
- Data preprocessing: Preprocessing data using tools such as:
  - StandardScaler: Used for normalizing input features.
  - SimpleImputer: Used for handling any missing values that may be present in a dataset.

Evaluation Metrics: Models will be evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC (Receiver Operating Characteristic - Area Under the Curve)

TensorFlow/PyTorch: In order to design, train, and deploy the AI models for both of the chatbots, TensorFlow/PyTorch will be utilized.

- TensorFlow:
  - For the creation and deployment of AI models for both.
  - The TensorFlow Serving API can be used for the deployment of the trained models for real-time interaction with the chatbot.
- PyTorch:
  - PyTorch can be used as an alternative if any issues with TensorFlow arise.
  - It is user-friendly and offers custom neural network architectures.

4. Pandas and NumPy: Pandas and NumPy will be used to handle and preprocess the symptom-based datasets and for performing data analysis for cancer risk scoring.

   Key Applications:

   - Cleaning, visualization, and transforming of raw symptom-based datasets.
   - Generating synthetic data if necessary to improve the model's performance.
   - Encoding categorical variables.
   - Exploratory data analysis (EDA) to view the trends and patterns that may arise in the data.

Natural Language Processing (NLP) Tools: In order to enable the general health query chatbot to respond efficiently to user input, that may vary in grammar, phrasing and spelling and for better accuracy, natural language processing must be used.

Tools:

- NLTK and SpaCy: NLTK and SpaCy are used for tokenization, lemmatization, and parsing of user input. It will also be utilised to analyse user input and extract important information inputted in the general chatbot.
- Hugging Face Transformers: To better understand users' queries and to promote a more natural flowing conversation, pre-trained models such as BERT or GPT can be used.

Database Management Systems:

- PostgreSQL or MySQL: In order to store patient information securely, a database management system such as PostgreSQL or MySQL must be used. Information such as:
  - User login information
  - Roles (such as patient, administrator)
  - Past cancer risk scores
- MongoDB (backup option):
  - If any issues arise with NoSQL, MongoDB can be used as an alternative to handle any structured/semi-unstructured data.

Cybersecurity Measures:

Cybersecurity is crucial for this project, as sensitive and confidential health data and login information will be handled. Cybersecurity requires the usage of tools, libraries, and frameworks. The key components and technologies that will be used to achieve this:

1. Encryption: In order to secure the data while it is being transmitted between the client and server (in transit) and while it is at rest (when the data is stored in the database).

   Technologies Used:

   (a) TLS/SSL Certificates: Using tools such as "Let's Encrypt" will enhance encryption, as it offers free and automated certificates. If advanced encryption is required, they also offer paid services.

   (b) Python Libraries:

i. pyOpenSSL and ssl: These libraries are crucial for enabling HTTP connections.

ii. Cryptography: This is a Python library used for encrypting confidential information such as user login credentials or past cancer risk assessments in the database.

iii. SQLAlchemy with AES encryption: Used for encrypting certain fields in the database.

2. Authentication: Secure login mechanisms, session handling, and multi-factor authentication (MFA) must be used to ensure that only authorized users can access their profiles and their data. It is also a crucial step to ensure that only authorized administrators are able to perform certain operations such as:

   - Deleting user accounts
   - Modifying data Technologies Used:

   (a) Password Hashing: Thanks to the passlib library that is available in Python, tools such as bcrypt, Argon2, or PBKDF2 can be used to securely hash users' passwords before storing this information in the database.

   (b) Session Management: Framework features such as Flask-Session for Flask or Django's built-in session middleware.

   (c) Multi-Factor Authentication (MFA):

   - Google Authenticator API or Twilio Authy: To enhance security and provide users with an additional layer of encryption, these tools can be used.
   - Pyotp (Python One-Time Password): A Python library that enables time-based one-time password (TOTP) generation.

3. Role-Based Access Control (RBAC): Role-Based Access Control ensures that users only have access to their own data. Administrators, on the other hand, are able to have wider control over the system.

   Technologies used:

   - Flask-Principal: A Flask extension used for controlling permissions and managing different user roles.

- Django Permissions (alternative approach): Django offers built-in features for assigning different permissions to different user roles.
- Database Role Management: Relational databases such as PostgreSQL/MySQL or NoSQL databases such as MongoDB allow for managing user roles and granting different permissions to these users.

4. Secure Coding Practices: In order to mitigate risks like SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF), secure coding practices must be used. This is done in order to prevent web vulnerabilities from occurring which can compromise sensitive user data. It is essential to sanitize and validate any and all data received from users before it is processed.

   Technologies Used:

   - Input Validation Libraries:
     (a) WTForms: This library in Flask is used to sanitize user input.
     (b) Regular expressions can be used in Python to also validate user input.
   - ORM Tools: These tools handle queries securely and in a parametrized way, preventing SQL injection.
   - CSRF Protection: Protection tools such as Flask-WTF or Django's built-in CSRF prevent malicious inputs from being submitted.

5. Logging and Monitoring: For the overall security of the platform, logging and monitoring are essential. This is to spot any malicious activity that may be taking place, such as login attacks (unauthorized access attempts), and to ensure that the system is operating smoothly.

   Technologies Used:

   - Logging Libraries:
     (a) Logging: Python offers a logging module that records user activities, errors, and any events out of the ordinary.
   - Monitoring Tools:
     (a) ELK Stack (Elasticsearch, Logstash, Kibana): Open-source tools used for real-time monitoring.

(b) Sentry: A service used to monitor errors and any performance issues that may arise.

6. Backup and Recovery: Regular backups are crucial to minimize any downtime and maintain data integrity. It is also required to ensure that in the event of an accident, confidential data can be recovered.

   Technologies Used:

   - PostgreSQL's pg dump or MySQL's mysqldump: For regular and scheduled backups.

7. Compliance with Legal Standards: In order to be transparent with users, prevent legal and ethical issues, and ensure the project complies with global and regional data protection laws, different technologies can be used.

   Technologies Used:

   - GDPR Compliance: Tools for the anonymization of user data and ensuring consent is received for data collection.
   - HIPAA Compliance: Encryption, secure access control, and audit trails for protecting health information.

4.2 Hardware Integration

Achieving a seamless development and deployment of the project depends on the successful integration of the hardware components. The role each hardware component plays is detailed below.

4.2.1 The Hardware Needed to Develop the Project (and to Support at Deployment Stage)

4.2.1.1 Development Stage

The hardware required during the development phase of this project is as follows:

- Development Machines (Laptops/Desktops):

  1. Processors: In order to handle the large computational workload of training the models and testing, multi-core processors are required (e.g., Intel Core i7 or AMD Ryzen 7).

2. RAM: For smooth data processing and machine learning operations, a minimum of 16GB of RAM is required.
3. Storage:
   - SSD (Solid State Drive): A minimum of 512GB of storage is required for quick data read/write operations.
   - A reasonable amount of space must be given to store symptom-based datasets, pre-trained NLP models, and machine learning checkpoints.
4. GPU (Graphics Processing Unit):
   - A dedicated GPU is necessary to speed up the testing and training process of models.
5. Operating System:
   - Such as Windows, macOS, or Linux.

- External Storage Devices: High-capacity external drives or cloud-based storage solutions for backing up datasets, trained models, and project files.
- Network Connectivity: To download large datasets and pre-trained models and to deploy models to the cloud, a high-speed and stable internet connection is required.

4.2.1.2 Deployment Stage

For the deployment stage, the hardware must be sufficient for delivering real-time interactions with the users and managing multiple requests from the users at the same time.

- Cloud Infrastructure: Cloud platforms such as Amazon Web Services (AWS), Google Cloud Platform (GCP), or Microsoft Azure will be used to host the web application to deploy the AI models.
  - Compute instances (e.g., AWS EC2 or GCP Compute Engine): Will operate the backend services and manage AI inference tasks.
  - Cloud storage solutions (e.g., AWS S3 or Azure Blob Storage): Will safely store user data and system logs.

- Server Types:

    1. CPU servers: To manage user data, user requests, and run the general health chatbot (simpler tasks).

    2. GPU servers: For more complex tasks, such as predicting cancer risk scores using deep learning models deployed with TensorFlow or PyTorch models.

- Database Server: Cloud databases like AWS RDS or Google Cloud SQL will be used to store sensitive user data such as login credentials and past risk scores safely.
  These servers offer secure and efficient data storage and retrieval.

- Load Balancer: This is a tool that distributes incoming user requests evenly across multiple servers. This will prevent the performance of the chatbots from slowing down when multiple users use it simultaneously.

4.2.2 High-Level System Architecture Overview

4.2.2.1 System Architecture

The system is composed of three parts:

1. **Frontend Layer**: This is the interface where users are able to log in and interact with the chatbots and view their past risk scores.

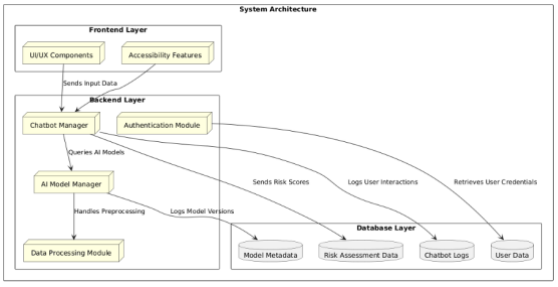


Figure 4.1: System Overview

2. **Backend Layer**: This layer is in charge of processing the data, hosting the AI models, and managing database interactions.

3. **Database Layer**: All of the user and system data is stored here.

4.2.2.2 Packages/Classes and Their Responsibilities

- **Frontend Layer**

  – Package: frontend –

  Components:

  1. UI/UX: Implements the user interface, including chatbot interaction screens, user login/signup forms, and result displays.
  2. Accessibility: Adjusts themes and font sizes for accessibility of users who may require them.

  – Responsibility:

  1. Communicates with the backend through APIs to send user inputs and receive responses.
  2. Renders cancer risk scores and general health advice from the backend.

- **Backend Layer**

  – Package: backend –

  Components:

  1. Chatbot Manager:
     - Directs user inputs to the correct chatbot (cancer risk assessor or general health advice chatbot).
     - Handles natural language processing tasks like tokenization and response generation.
  2. AI Model Manager:
     - Hosts the trained AI models for cancer risk evaluation and general health query responses.
     - Loads and deploys models using TensorFlow or PyTorch.
  3. Authentication Module:
     - Manages user authentication through the implementation of safe methods such as password hashing and Multi-Factor Authentication (MFA).

4. Data Processing Module:
    * Preprocesses user-provided data such as symptom-based inputs.
    * Handles missing values, normalization, and feature engineering.

– Responsibility:

1. Ensures that the data transmission between the frontend and backend is secure.
2. Executes AI models and returns results.
3. Controls and enforces user roles (e.g., users vs administrator).

- **Database Layer**

– Package: database – Components:

1. User Data: Stores user credentials, roles, and authentication logs.
2. Risk Assessment Data: Stores previous cancer risk assessment scores.
3. Model Metadata: Tracks AI model versions and performance metrics.
4. Chatbot Interaction Logs: Logs interactions users have with the chatbot for future debugging and analysis.

– Responsibility:

1. Stores all data securely and efficiently.
2. Serves up data in real-time for user dashboards and AI interactions.

4.2.2.3 High-Level View of the Database

The database will consist of a hierarchical structure. This is to manage relationships between different data entities. The structure will be as follows:

Table 4.1: High-Level View of the Database

4.2.3 Functional Requirements and System Analysis

4.2.3.1 Use Case: Sign Up

Table Name	Attributes
Users	user id (PK), name, email, password hash, role, created at, last login
Roles	role id (PK), role name, permissions
Cancer Risk Scores	risk _id (PK), user id (FK), cancer type, risk score, assessment date, symptom
Symptoms	symptom id (PK), cancer type, symptom name, weight
Chatbot Interactions	interaction id (PK), user id (FK), query, response, timestamp
Models	model id (PK), model name, version, metrics, created at, deployed at

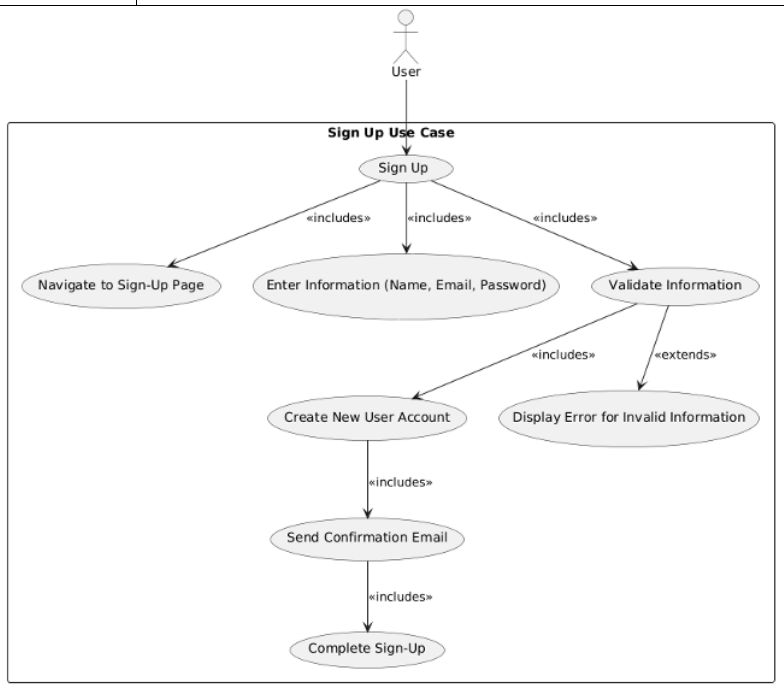


Figure 4.2: A sign up Use Case [**?**]

- Actors: User
- Goal: Register a new account on the website.
- Precondition: The user is not logged in & does not have a pre-existing account.
- Main Flow:

1. User navigates to the page where prompted to log in/sign up.

2. User clicks signup.
3. User enters required information (name, email, password).
4. System validates the information.
5. System creates a new user account and sends confirmation email.
6. User completes sign-up.

- Alternative Flow:
  - Invalid Information: System returns an error message for incorrect or missing values inputted by the user.

4.2.3.2 Use Case: Login

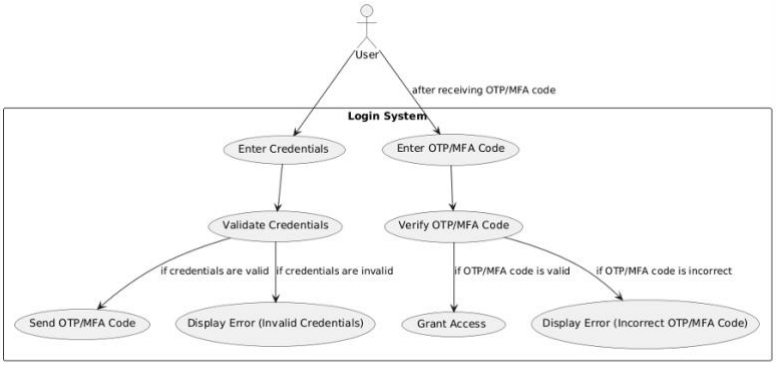


Figure 4.3: A login use case for the system.

- Actors: User
- Goal: Log in and gain access to their dashboard.
- Precondition: User already has a pre-existing account.
- Main Flow:
  1. User enters their email and password into the login form.
  2. System validates information.
  3. If credentials are validated successfully, the system sends a One-Time Password (OTP) or uses an MFA app (e.g., Google Authenticator) to generate a code.
  4. System verifies OTP/MFA code.

5. System grants access to the user's dashboard.

- Alternative Flow:
  - Invalid Credentials: System displays an error and prompts the user to reenter information.
  - Incorrect OTP/MFA Code: System displays an error and prompts the user to enter the correct MFA code.

4.2.3.3 Use Case: Health Chatbot Interaction

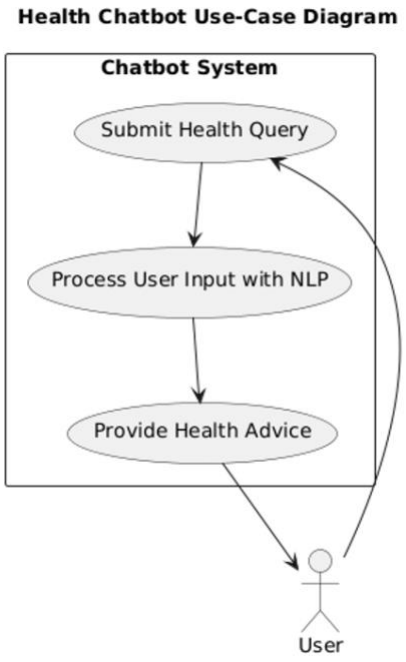


Figure 4.4: A general health chatbot interaction use case for the system.

- Actor: User
- Goal: Receive general health advice from the chatbot.
- Scenario:
  1. The user asks a health-related question.
  2. The chatbot processes the input by using NLP models.
  3. The system returns general non-diagnostic health advice to the user.

- Challenges: Understanding diverse grammar, phrasing, and spelling inputted by the user and offering accurate advice.

4.2.3.4 Use Case: Adjust Font Size

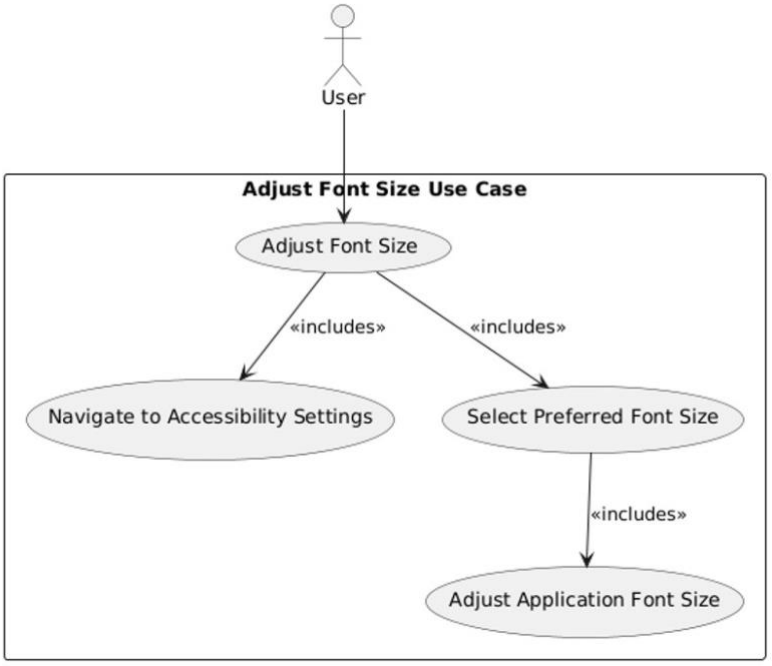


Figure 4.5: Font Size Adjustment Feature

- Actors: User
- Goal: Change the font size to accommodate users with disabilities.
- Precondition: User is logged in.
- Main Flow:

    1. User navigates to the accessibility settings.
    2. User adjusts the font size to what they prefer.
    3. System adjusts the font size for the website.

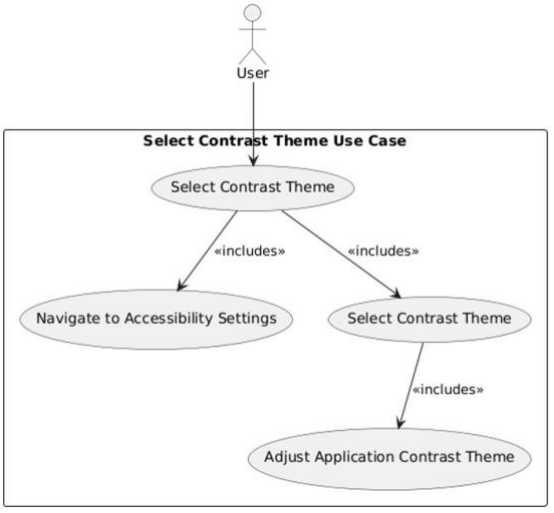


Figure 4.6: High-Contrast Theme Feature

4.2.3.5 Use Case: Switch to High-Contrast Theme

- Actors: User
- Goal: Switch to a high-contrast theme for readability.
- Precondition: User is logged in.
- Main Flow:
    1. User navigates to the accessibility settings and selects the ”High-Contrast Theme.”
    2. System switches the current theme to the selected high-contrast theme.

4.2.3.6 Use Case: Cancer Risk Assessment

- Actor: User
- Goal: Assess cancer risk based on symptoms provided by the user.
- Scenario:
    1. The user logs in.
    2. The user provides symptoms through a series of questions.
    3. The input provided by the user is processed using a machine learning model.
    4. A risk score is calculated and provided to the user.

- Challenges: Collecting sufficient data required for accurate risk predictions.

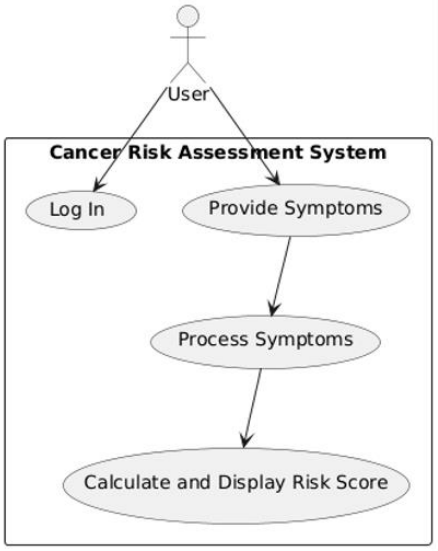


Figure 4.7: Cancer Risk Chatbot

4.2.4 Data Flow Diagrams (DFDs)

4.2.4.1 Level 0 DFD: Context Diagram

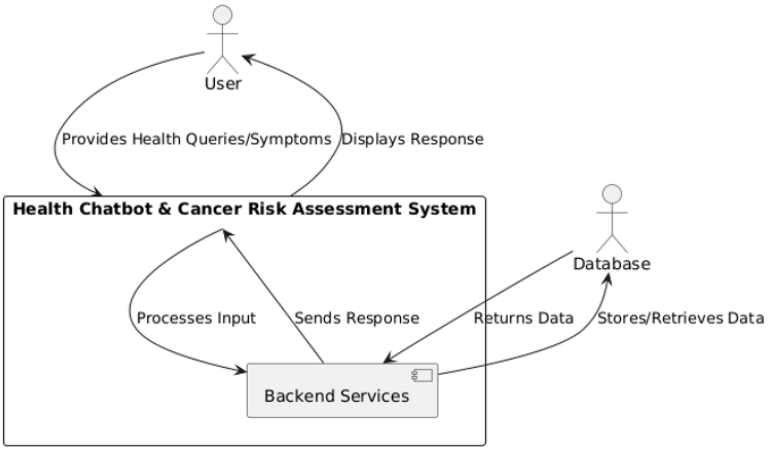


Figure 4.8: Level 0 Data Flow Diagram

- Actors: Users, Backend, Database.
- System: The health chatbot and cancer risk assessment system.
- Description: Shows data provided by the users, which are processed by the backend services, and then stored/retrieved from the database.

4.2.4.2 Level 1 DFD: Deeper Flow

- Process Descriptions:

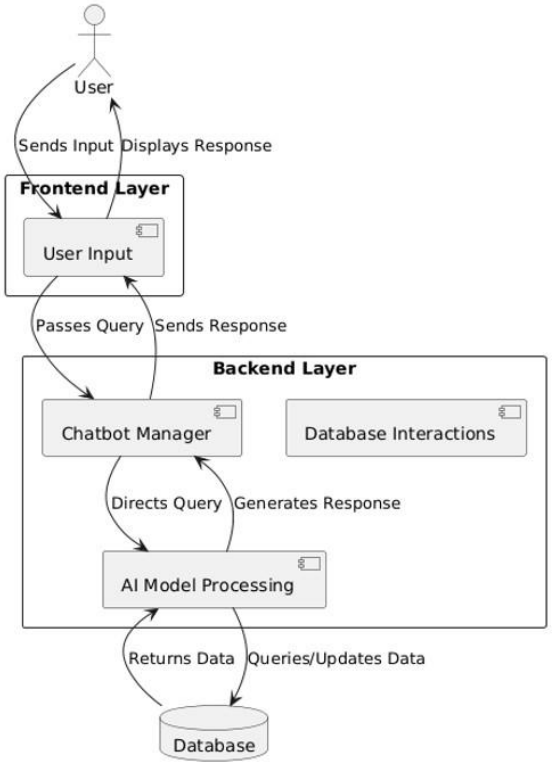


Figure 4.9: Level 1 Data Flow Diagram

  - User Input: The frontend interface sends the user input to the backend.
  - Chatbot Manager: Directs the flow based on the nature of the query.
  - AI Model Processing: Generates general health advice and responds to the user (for general queries) or provides cancer risk scores.
  - Database Interactions: Stores user data and cancer risk assessment scores.

4.3 Risk Assessment

This section will discuss the potential risks that may affect the successful completion of this project. It will additionally discuss, potential mitigation strategies for these risks. A good approach to assessing these risks are basing them on their severity and probability of occurrence. A breakdown of each risk is shown below:

- **Performance**

- Issue: Latency during real-time chatbot interaction.
- Severity: Medium
- Probability: Medium
- Mitigation: Use caching, optimization in model processing, and possibly use lightweight alternatives.

- **Data Quality**
  - Issue: Insufficient symptom-based datasets.
  - Severity: High
  - Probability: High
  - Mitigation: Keep updated on newly released datasets; use synthetic data generation methods to fill gaps. Review dataset quality regularly.

- **Security**
  - Issue: Data breaches of confidential user information.
  - Severity: High
  - Probability: High
  - Mitigation: Implement TLS/SSL encryption, secure coding practices, session management, and MFA to secure user data.

- **Data Storage**
  - Issue: Issues related to storing large amounts of data.
  - Severity: Medium

- Probability: Medium
- Mitigation: Use cloud storage solutions that provide scalability options like AWS S3 or Azure Blob Storage. Schedule regular backups.

- **Model Accuracy**
  - Issue: ML model underfitting or overfitting.
  - Severity: High
  - Probability: High
  - Mitigation: Apply hyperparameter tuning. Perform extensive cross-validation and test data assessments.

- **Third-party Tools/API**
  - Issue: If APIs or NLP authentications have service issues, it could cause delays in the project and make the platform unavailable to users.
  - Severity: Low
  - Probability: Medium
  - Mitigation: Design fallback solutions. Ensure APIs are updated regularly. Adopt alternatives in case one tool fails.

- **Timeline Management**
  - Issue: Unforeseen delays in project milestones.
  - Severity: Medium
  - Probability: High

- Mitigation: Use an iterative development approach; allocate buffer times for unforeseen circumstances. Use tools such as Trello or Gantt charts to track progress continuously.

4.4 Methodology

4.4.1 Research Approach

A systematic literature review In the domains of health informatics, AI-driven cancer risk detection, and conversational AI will be undertaken to perform the needed research.
The research will be conducted as follows:

1. **Literature Review:**
    - Use platforms such as PubMed and Google Scholar to find relevant research papers, books, articles and already existing projects.
    - Summarize the state-of-the-art advancements and key methodologies currently used in medical chatbots.
    - Analyse existing systems and models to understand how they work and to analyse areas where further improvements can be made.
2. **Identification of Gaps:**
    - Identify the existing gaps in AI-driven cancer risk assessment tools and health informatics solutions.
    - Understand and document the limitations and challenges that need to be overcome by the proposed project.
3. **Documentation:**
    - Document the findings from literature reviews .
    - Keep detailed notes.

4.4.2 Skill Setup and Learning Approach

The project involves several technologies and tools that I need to be proficient in. This section details my approach to enhancing my skills:

1. **New Technology Training:**

    - TensorFlow and PyTorch: I have the necessary skills in this area, however there is room for improvement. Online platforms such as Coursera and Udemy offer courses on these topics. I will also examine the official documentation to use TensorFlow and PyTorch as effectively as possible to train and deploy models.

    - Flask and Django: I am comfortable with coding in Python however I will aim to further improve my web development skills by learning Flask and Django frameworks through reading official documentation and watching tutorials. Flask will be my main area of interest and hence it will prioritized.
      Django will be used if there is a need for robustness.

    - Natural Language Processing (NLP): Having a foundation in tools like NLTK, SpaCy, and Hugging Face Transformers is necessary for the implementation of chatbots. Tokenization, lemmatization, and text classification will be practiced through following documentation, developing small projects, and experimenting with pre-trained models.

    - Database Systems: I will improve on my database management skills by working on data handling exercises involving CRUD operations, database design, and user authentication.

2. **Skill Development Plan:**

    - Online Learning Resources: Utilize online learning platforms such as Coursera, Udemy, and YouTube. This will include working through projects to get hands-on experience with new technologies.

    - Official Documentation and Books: To understand the technologies and use them as effectively as possibly, read through books that specialise in these areas, and read through official documentation for each technology.

- Practice Projects: Implement practice projects such as building simple chatbots, machine learning models, and CRUD-based applications to apply newly learned concepts.

3. **Prototyping and Testing:**

   - As I navigate through the learning process, I will create prototypes of technologies. This will help me to demonstrate my understanding of each technology.

4.4.3 Project Management Approach

The project will follow a hybrid approach, both Waterfall and Agile methodologies will be combined, considering the nature of the project and the importance of sequential development phases while still allowing for some flexibility.

1. **Waterfall Approach:**

   - **Initial Phases:**

     – Conducting the background research, defining project requirements, and developing the system architecture will follow a sequential, Waterfall approach. This will help in providing a clear and structured basis for the project.

2. **Agile/Scrum Approach:**

   - **Development and Implementation Phases:**

     – An iterative approach will be adopted for the development of the chatbots,the web application and the database. Using a Scrum methodology, each component will be developed in ”sprints” which will last approximately 2-3 weeks.

     – The Scrum methodology will allow for flexibility and will allow for refinements and improvements.

     – A product backlog containing all tasks and features that need to be implemented will be kept. The work needed to be completed will be

chosen carefully at the start of every sprint and time efficiency will be prioritised.

3. **Tools for Project Management:**

    - Trello or Gantt charts will be used to monitor tasks, track deadlines, and to monitor the project's overall progress.
    - GitHub will be used for code management and version control.

4. **Risk Management:**

    - After each sprint, the sprints performance will be reviewed and the risks of the project will be documented.
    - The documentation will help to identify issues and will therefore allow me to take action.

4.5 Implementation Plan Schedule

The developmental phase of this project will be divided into several phases to ensure incremental development and timely completion of each stage. This plan allows for the project to be completed in time. The breakdown of the plan schedule is as follows:

- **Phase 1: Data Collection and Preparation (Weeks 1-2, Semester 2)**
    - Finalize the symptom-based datasets to be used for cancer risk assessment.
    - Use synthetic data generation methods if an appropriate dataset can not be found.
    - Perform data cleaning and augmentation,

  - Validate the datasets to ensure high quality for training models.

- **Phase 2: Machine Learning Model Development (Weeks 3-5, Semester 2)**
  - Using the datasets, train and fine-tune machine learning models for cancer risk prediction.
  - Evaluate models using accuracy, F1 score, and ROC-AUC metrics.
  - To improve the prediction accuracy of models, try alternative models (e.g., Random Forest, Logistic Regression).

- **Phase 3: NLP Model Fine-Tuning and Integration (Weeks 4-6, Semester 2)**
  - Fine-tune BERT or GPT models for health-related queries.
  - Handle different phrasing, grammar and misspellings by users by integrating error correction mechanisms.
  - Perform initial testing of the NLP chatbot functionality with mock data.

- **Phase 4: Backend and Frontend Integration (Weeks 7-9, Semester 2)**
  - Develop APIs to connect the NLP and machine learning components with the web application.
  - Develop the frontend interface for both chatbots, the focus should be user-friendliness and accessibility.
  - Set up and integrate the database for the management and storage of user data.

- **Phase 5: Testing and Optimization (Weeks 9-11, Semester 2)**
  - Ensure seamless integration has been achieved by conducting testing.
  - Optimize performance by addressing any existing latency issues.
  - Incorporate a feedback loop to enhance chatbot responses and functionalities.

–

- **Phase 6: Deployment and Finalization (Weeks 12-13, Semester 2)**
  - Deploy the system on a cloud platform (e.g., AWS EC2).

    Finalize the required documentation.
  - Conduct a final presentation of the project.

Buffer time: 1 week, Semester 2)

- In the event of delays/ unforeseen circumstances, allocate buffer time.

4.6 Evaluation

To determine the success of the project, a systematic approach to evaluation will be taken. This will include both qualitative and quantitative measures to evaluate how well the functional requirements have been met as well as the problem solving approach that has been taken.

- **Functional Requirements Evaluation:**
  - Measure the accuracy of the cancer risk assessment chatbots by using metrics such as accuracy, precision, recall, F1 score, and ROC-AUC .
  - Confirm that the general health chatbot is able to understand and process handle user inputs that may vary in grammar, phrasing and spelling.
- **User Experience Evaluation:**
  - Conduct user testing sessions.
  - collect feedback on the interface, accessibility, and ease of use.
  - Ensure accessibility features such as contrast themes and adjustable font sizes operate effortlessly for users who may require them.
- **Overall Problem-Solving Evaluation:**

  -
  - Analyse the system's functionality and compare it with the original project goals to ensure goals have been met.
  - Assess whether both the general health chatbot and the cancer risk assessment chatbot are able to offer health support accurately to users.

- **Quantitative Success Metrics:**

  Achieve an F1 score of at least 0.85 for cancer risk prediction models.

  - Ensure 90% of user input in the general health chatbot are accurately understood, processed and responded to.
  - Maintain as efficient response times as possible for both chatbots.

4.6.1 Prototype Implementation for Chatbot Functionality

4.6.1.1 Cancer Risk Assessment

For the purpose of demonstrating the functionality of the cancer risk chatbot, an API endpoint was built and tested for lung cancer using Postman. The main objective of this prototype was to validate the logical flow of the chatbot's questioning and its ability to respond to user input.

Objective:

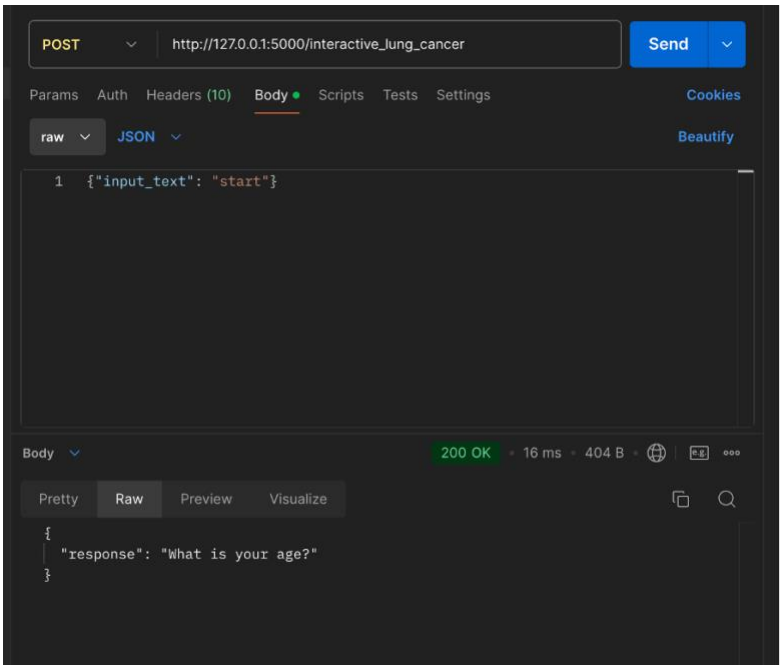
1. Develop an interactive questioning mechanism that assesses the risk of lung cancer.
2. Confirm that its ability process user input dynamically to predefined health questions.
3. Test the backend logic for conditional question flow.

Implementation: A Flask-based API endpoint (/interactive_lung cancer) was created to handle the interaction flow. This endpoint receives input from users in JSON format. It processes it and asks the next question immediately. Here is a description of how it works:

1. User begins the process by typing "begin".

2. The user responds to the questions asked (e.g., ”Do you experience frequent colds?”).
3. Processing: The input is processed through the backend and asks the next question, which is based on a pre-defined question tree.
4. The system responds in JSON format and continues to ask the questions.

Example workflow:



1. User starts session.

Figure 4.10: Caption for start



2. Model asks its initial question.

Figure 4.11: Caption for nextq

3. Model asks next question.

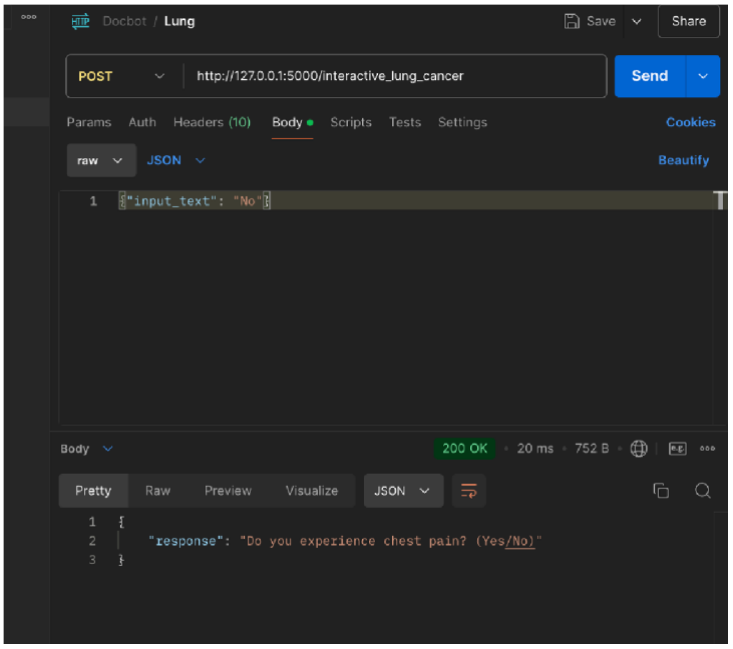


Figure 4.12: Next question

4. Example of questions asked.

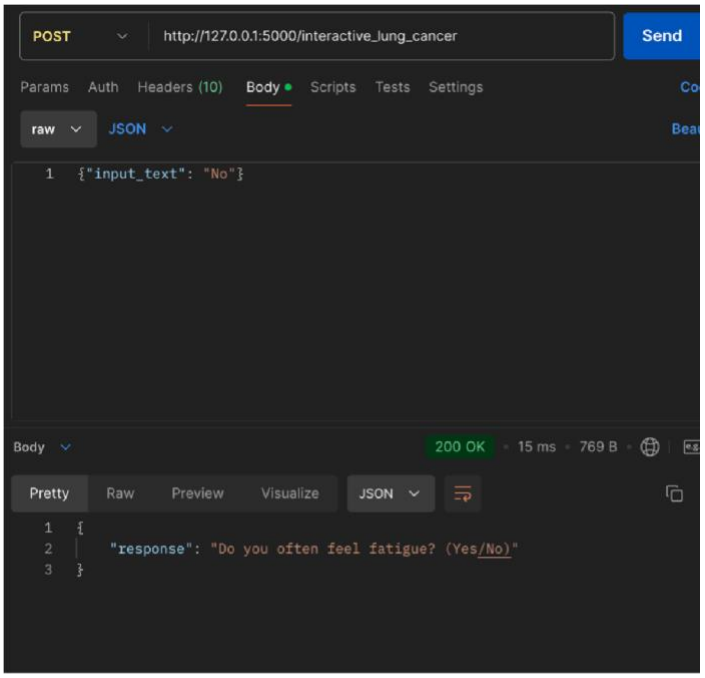


Figure 4.13: Next question

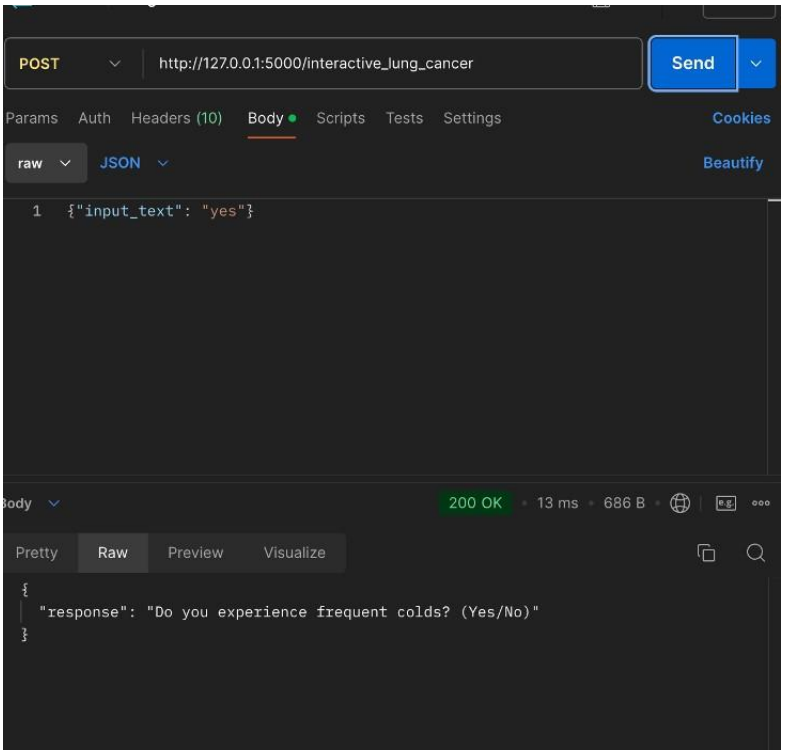


Figure 4.14: Next question

5. Finally, the model provides a risk score.

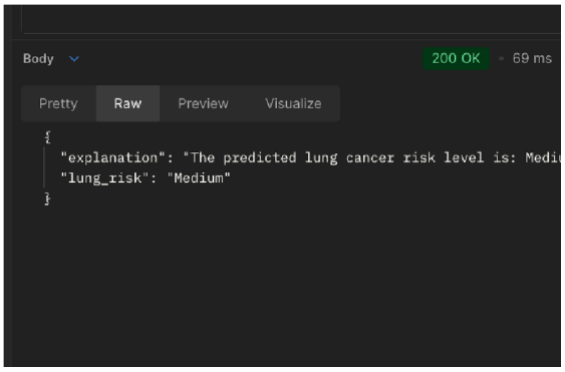


Figure 4.15: Risk score

This prototype validates the following set of criteria:

1. Interactivity: Chatbot can handle the set of questions seamlessly.
2. Backend logic: The system is able to process user inputs and select the next question.
3. API functionality: The Flask endpoint handles requests and responses without encountering any issues.
4. Risk Score: The model provides a risk score succesfully based on the answers provided by the user.

Next Steps:

1. Front-End Integration: Integrating the backend functionality into a user-friendly interface.
2. Expanded Questions: Expanding the set of questions asked to include other health issues that may be relevant and therefore enhance the accuracy.
3. Repeat this process for other cancers: Repeat these steps for the other cancers that will be covered (breast cancer, skin cancer, prostate cancer, cervical cancer).

4.6.1.2 General Health Chatbot

In the prototyping stage of the project, I developed a basic version of the general health query chatbot. This chatbot utilizes OpenAI's language model in conjunction with Natural Language Processing (NLP) techniques to interact with users and provide healthrelated advice based on the symptoms they describe.

Prototype Features:

1. **Basic User Interaction:**
    - The chatbot greets users and offers help for health-related queries. For example, when a user says, "I have a headache," the chatbot responds with questions to understand the user's condition better, such as asking about the type of headache, its location, and the duration.
2. **NLP Integration:**
    - I used NLP tools to extract relevant health symptoms from the user's input and generate follow-up questions or recommendations. This helps in making the conversation more natural and relevant to the user's needs.
3. **Use-Case Example:**
    - In a scenario where a user mentions they have a headache, the chatbot asks targeted questions like:
        - Type of Headache: Is it dull, throbbing, or sharp?

  - Location: Where is the headache located?
  - Duration: How long have you had the headache?

- Based on these answers, the chatbot offers tailored advice such as staying hydrated, resting in a quiet room, or considering over-the-counter pain relievers.

Objective:

- The primary goal of the prototype was to validate the interaction flow of the chatbot and ensure that it can respond to user inputs in a helpful manner. Additionally, this prototype allowed me to explore how the integration of OpenAI's capabilities with basic NLP tools could help in developing a health support system.

Screenshots

- Screenshots of the chatbot conversation (such as the ones attached below) are used to demonstrate the prototype functionality. The interaction showcases the bot's ability to understand user input and respond with detailed follow-up questions to gather further information.

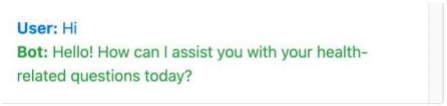


Figure 4.16: Screenshot 1 of the chatbot conversation.

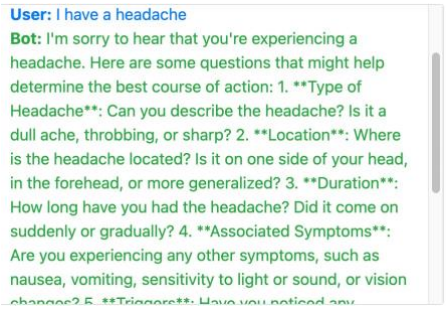


Figure 4.17: Screenshot 2 of the chatbot conversation.

Chapter 5

Conclusions and Future Work

5.1 Discussion

Several challenges were faced during this phase of the problem. Some of these challenges include:

- **Data Collection Issues:** Obtaining symptom-based datasets was quite challenging, most of the datasets for each of the cancers to be assessed were based on lab results / scans. High-quality datasets are required to achieve the most optimal predictions. The lack of high-quality data can result in false positives/negatives, which can be harmful for the users of the platform.

  Future Approaches to Address This Challenge:

  - **Synthetic Data Generation:** Create symptom-based datasets with the help of synthetic data generation methods. Tools such as GANs (Generative Adversarial Networks) or simulation-based approaches can mimic the characteristics of real data.
  - **Data Augmentation:** To diversify the current datasets, use data augmentation techniques. For example, reformatting datasets to emphasize symptom descriptions or adding noise to simulate variability in user-reported data.
  - **Open-Source Data Exploration:** Keep updated on platforms such as Kaggle, UCI Machine Learning Repository, or government health portals for symptom-based datasets that may be published.
  - **Iterative Data Improvement:** Consistently refining and validating datasets as more data becomes available can help the chatbots accuracy to improve.

  - **Customizing NLP Models for Domain-Specific Queries:** Although BERT and GPT are exceptional tools for this project, applying these models to general

health chatbot queries presents challenges. The chatbot must be able to understand and process user inputs that may contain grammatical errors, misspelling of words, variation in phrasing etc..

A notable amount of effort and resources is required to fine-tune these models to help them understand, process, and respond to such inputs.

Future Approaches to Address This Challenge:

- **Fine-Tuning Pre-Trained Models:** Adapt models like BERT to the specific context of health-related queries through domain-specific fine-tuning using labeled health datasets.
- **Real-World User Feedback:** User inputs that may be misspelled, grammatically incorrect, or phrased incorrectly can be collected using a feedback loop. These collected inputs can help improve the chatbot's natural language understanding capabilities.
- **Error Detection and Correction:** Pinpoint and correct the input errors in real-time before processing it by using extra NLP layers.

- **Time Constraints:**
  - **Description:** Given the project's complexity and its use of various technologies at once (developing machine learning models, integrating NLP tools, and building the web application) research time was a challenge during this phase. Balancing coursework with research was sometimes difficult.
  - **Future Impact:** Time constraints could decrease the quality of the developed components and could result in incomplete testing. This would be detrimental to the project as it would impact the reliability of the system.
  - **Future Approach:**
    - Prioritize core functionalities (e.g., chatbot responses and cancer risk evaluation) to ensure that critical components are fully functional.
    - Follow an iterative development approach. Focus on building the basic versions of the core components and then improving these basic versions gradually.
    - Set aside buffer time for unforeseen circumstances/delays and Set realistic milestones.

Chapter 6

Conclusions and Future Work

6.1 Conclusion

This phase of the project focused on identifying key challenges, exploring solutions, and establishing a foundational understanding of the system architecture and technologies required. The major findings and conclusions are as follows:

- **Background:** Research has shown that AI is currently being adopted globally for medical purposes. It aims to benefit the field of medicine greatly. Both the general and cancer risk assessment chatbot can provide accessible and preliminary health support.

  Using technologies like machine learning, Natural Language Processing (NLP), and web frameworks was identified as essential for building the proposed system.

- **Problem Description:** Two major components were identified for development:

  1. A *Cancer Risk Assessment Chatbot* that uses machine learning models to analyze symptoms inputted by users and return a risk score.
  2. A *General Health Chatbot* processes medical queries inputted by users and provides non-diagnostic medical advice

  Acquiring symptom-based datasets, learning NLP customization, and time constraints were the main difficulties faced during this phase of the project. These challenges have resulted in a shift of strategy moving forward in order to succesfully achieve the aims for this project.

- **Solution Approach:**

  - Addressing *Data Collection Issues* will involve generating synthetic data, augmenting datasets, and exploring open-source repositories to ensure robust model training.

- For the *General Health Chatbot*, fine-tuning pre-trained models such as BERT and implementing error detection mechanisms were identified as critical for handling domain-specific user inputs.
- Managing *Time Constraints* will consist of an iterative developmental process, core functionalities will be prioritized and buffer time will be set aside to cover any unforeseen delays that may arise.

This phase has helped develop a clear understanding of the projects scope, challenges and solutions. A strong foundation has been laid for the development phase.

6.2 Future Work

A few tasks remain to be addressed to ensure the success of this project. These tasks include:

- **Complete Symptom-based Dataset Acquisition:**
  - Explore and acquire more symptom-based datasets for the cancers that will be included in the chatbot (lung cancer, prostate cancer, cervical cancer, breast cancer, skin cancer)
  - If symptom-based datasets are not acquired, generate data synthetically.
- **Model Development and Training:**
  - Using the datasets, train machine learning models for cancer risk assessment.
  - Fine-tune pre-trained NLP models (e.g., BERT or GPT) for the general health chatbot.
- **Integration of System Components:**
  - Ensure seamless integration between each component.
  - Ensure strong communication between components with the use of APIs.
- **Testing and Validation:**

- Ensure reliability and accuracy of chatbot by consistent and thorough testing.

- **User Interface Improvements:**

  - Develop a professional yet easy to use platform, the users should be able to navigate through the platform easily regardless of their technical background.
  - Ensure accessibility features such as contrast themes and adjustable font sizes are available for users with disabilities.

- **User Feedback Mechanism:**

  - Improve accuracy and performance of the chatbots by implementing a feedback loop which will collect user input and responses.

- **Addressing Time Constraints:**

  - Set realistic goals and establish an achievable timeline for implementation, testing, and deployment phases.
  - Allow for adjustments by adopting agile methodologies.

Addressing these tasks will help the project transform into a fully functional system that is highly accurate and accessible.

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Appendix A Code

Snippets

Put appendix material in this section e.g. code snippets

USE THE APPENDICES

Appendix B

Wireframe Models